

KWABENA ADU AGYEMANG

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EDUCATION

Doctor of Philosophy, Computer Science
New Mexico State University

August 2025 –
New Mexico, USA

Master of Science, Computer Science
New Mexico State University

August 2023 – August 2025
New Mexico, USA

Master's Thesis: Understanding Student Practices for Using AI-Powered Learning Assistants: Ethical and Effective Strategies for Academic Success

Bachelor of Science, Computer Science
Valley View University

October 2017 - December 2021
Accra, Ghana

Undergraduate Thesis: Robust Ensembled Algorithm for Network Intrusion Detection System

RESEARCH EXPERIENCE

New Mexico State University
Participatory Live Experiences (PLEX) Lab
Advisor: Dr. Bill Hamilton

New Mexico, USA
August 2024 - Present

- Researching human experiences in emerging live media technologies, including real-time video, text chat, and collaborative tools.
- Collaborating on LiveMâché, a web-based app for free-form curation, supporting ideation through creative strategies like Collect, Assemble, and Exhibit.
- Investigating spatial organization patterns to design tools that enhance collaboration and creative relationships.
- Assisting with technical development, user testing, and integrating findings into my master's thesis.

New Mexico State University
School of Nursing
Supervisor: Jochebed Bea Ade-Oshifogun, Ph.D., Associate Professor of Nursing

New Mexico, USA
September 2023 - Present

- Assisting in three research projects on intimate partner abuse, nursing education, and marital dissatisfaction, contributing to manuscripts ready for submission.
- Conducts qualitative data analysis using NVivo and statistical analysis with SPSS.
- Collaborates on questionnaire design, data collection, and manuscript preparation.
- Synthesize findings to address cultural, societal, and educational challenges.

Valley View University
Artificial Intelligence & Machine Learning Lab
Advisor: John Kingsley Arthur

Accra, Ghana
September 2020- October 2021

- Conducted research on machine learning applications in cybersecurity, focusing on a Network Intrusion Detection System.
- Developed an Autoencoder-based framework for feature selection and applied classification algorithms to improve detection accuracy.
- Addressed challenges like false alarms, detection time, and processing complexity.
- Completed an undergraduate thesis on "Robust Ensembled Algorithm for Network Intrusion Detection Systems" under his supervision.

TEACHING EXPERIENCE

New Mexico State University
Teaching Assistant
Department (Computer Science)

New Mexico, USA
January – May 2025

CS 382 /CS 532 Modern Web Technologies

Spring 2025

COSC124 Procedural Programming (C++)	Fall 2021
COSC117 Fundamentals of Computing Skills	Fall & Spring - 2021, 2022, 2023
COSC113 Elements of Programming	Fall & Spring - 2021, 2022
INFT115 Information Technology Foundation I	Fall & Spring - 2021, 2022, 2023
INFT116 Information Technology Foundation II	Fall 2022 & Spring 2023
INFT352 Enterprise Information Security	Summer 2022
RNUR101 Nursing and Midwifery Informatics	Fall & Spring - 2021, 2022, 2023
COSC255 Operating System	Summer 2022
INFT360 Web Engineering & Application System	Spring & Summer 2023
COSC240 Systems Programming	Spring 2023
INFT346 DotNet Programming	Spring 2023
INFT410 Internet Enabled Application Systems Development	Spring 2023
COSC224 Object-Oriented Programming	Fall 2022
INFT280 Management Information Systems	Spring 2023
BBIS213 Introduction to Business Information Systems	Spring 2023
COSC280 Information Systems	Spring 2023

RESEARCH PUBLICATIONS

- “Development of a Culturally Sensitive Questionnaire to Measure Intimate Partner Violence (IPV) Risks Among First-Generation Latino Immigrants in Southern New Mexico”
Authors: Bosede Ade-Oshifogun, Ph.D., Jodi Patterson Ph.D, Kwabena Adu Agyemang, BSc,
Status: Work-in-progress
Description: The study aims to develop a culturally sensitive questionnaire to assess intimate partner violence (IPV) risks among first-generation Latino immigrants in Southern New Mexico. Using focus groups and a phenomenological approach, it will identify IPV risk factors and validate an instrument for early intervention. Results will support targeted therapy and inform broader IPV prevention strategies.
- “Intimate Partner Abuse Among Nigerian Nurses: A Comparative Study Between USA and Nigeria”
Authors: Bosede Ade-Oshifogun, Ph.D., Ngozi Mbibi, DNP, Josephine Ezemobi, DNP, Kwabena Adu Agyemang, BSc
Status: Under Review
Description: This study explores the occurrence of intimate partner abuse among Nigerian nurses in the USA and Nigeria. It examines the relationships between demographic variables and abuse scores, providing insights into cultural and societal influences on the prevalence of intimate partner abuse.
- “Nursing students’ perception of the nursing educational environment at Andrews University.”
Authors: Jochebed Bea Ade-Oshifogun, Ph.D., Kwabena Adu Agyemang, BSc., Dr. Temitope O Ade-Oshifogun
Status: Under Review
Description: This study analyzes nursing students' perceptions of their educational environment using the Dundee Ready Educational Environment Measure (DREEM) questionnaire. The research evaluates five key DREEM domains and compares these perceptions across sophomore, junior, and senior cohorts.
- “Perceived Causes of Marital Dissatisfaction among Nigerian Immigrants in North America: A Qualitative Exploration”
Authors: Jochebed Bea Ade-Oshifogun, Ph.D., Augusta Olaore, PhD. Kwabena Adu Agyemang, BSc.
Status: Published, 2025
Description: This qualitative study explores the causes of marital dissatisfaction among Nigerian immigrants in North America, aiming to inform culturally sensitive interventions to enhance marital satisfaction and prevent intimate partner violence.

ACADEMIC PRESENTATIONS

- “Intimate Partner Abuse Among Nigerian Nurses: A Comparative Study Between USA and Nigeria”
4th Annual National Association of Nigerian Nurses In North America (NANNNA) Scientific Conference On Global Health, North Carolina - November 7-9, 2024
- “FC-IWO-RDBN: A Novel Robust Framework for PDF Malware Detection and Classification”
SMARTBLOCK4AFRICA 2024 Conference, Valley View University - September 30, 2024
- “An Improved Machine Learning-based Intrusion Detection System”
ICT EXPO, Valley View University - January 13, 2022
- "Robust Ensembled Algorithm for Network Intrusion Detection Systems,"
Departmental Defense, Valley View University - October 13, 2021

WORK HISTORY

- Graduate (Teaching) Assistant** January 2025 – Present
New Mexico State University New Mexico, USA
- Assisted in delivering lectures, grading assignments, and providing technical support to students in HTML, CSS, JavaScript, and responsive web development frameworks.
- Graduate (Research) Assistant** September 2023 – Present
New Mexico State University New Mexico, USA
- Conduct research, data analysis, and literature reviews to support ongoing academic projects.
 - Assist in experiments, manuscript preparation, and research dissemination through presentations.
- Teaching Assistant** November 2021 – August 2023
Valley View University Accra, Ghana
- Conducted tutorials, managed grading, and developed instructional materials for IT-related courses.
 - Provided administrative and academic support to enhance student learning outcomes.
- Web Development Instructor (Part-time)** January 2023 – March 2023
New Nation School Accra, Ghana
- Taught HTML, CSS, and JavaScript, fostering hands-on learning through project-based activities.
- IT Internships**
- Valley View University – Information Technology Services** May 2021 – July 2021
- Komfo Anokye Teaching Hospital – IT Unit** May 2019 – July 2019
- MTN Ghana – Information Systems Division** June 2018 – August 2018
- Supported IT infrastructure projects, including network troubleshooting, hardware repairs, and software installations.
 - Conducted data analysis and improved system performance across various roles.
 - Developed teamwork and technical skills in high-pressure professional environments.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Java, JavaScript, C#, R, PHP
- **Database:** MySQL, MongoDB, SQLite, Microsoft SQL Server
- **Tools & Technologies:** Git/GitHub, Docker, Kubernetes, Visual Studio Code, Jupyter Notebook
- **Frameworks & Libraries:** TensorFlow, PyTorch, Flask, React, Angular, Keras, Scikit-learn
- **Game Development:** Unity, Unreal Engine, Godot
- **Web Development:** HTML5/CSS3, JavaScript (Node.js, Express.js), PHP, WordPress
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn
- **Operating System:** Windows, Linux, MacOS

ACTIVITIES

Professional & Academic Affiliations

- The Society for Collegiate Leadership & Achievement – Member
- Phi Sigma Pi National Honor Fraternity - Member
- Computer Science Graduate Student Organization - Member
- Ghana Students Association, New Mexico State University – Member
- Computing Sciences Students Association, Valley View University - Member

Leadership Experience

- Computing Sciences Students Association, Valley View University
Member, Advisory Board, 2020-2021
- Students' Representative Council, Valley View University
Member, Advisory Board, 2020 – 2021

Public Speaking

- *Data Analytics using Python and Machine Learning Models*, Valley View University
- *Information Technology in Today's World*, Valley View Basic School

May 2022
January 2022

Honors & Awards

- Most Influential Student, 2021
Students' Representative Council, Valley View University
- Student of the Year, 2021
Students' Representative Council, Valley View University